Doubly Stochastic Mean-Shift Clustering

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Abstract

Standard Mean-Shift algorithms are notoriously sensitive to the bandwidth hyperparameter, particularly in data-scarce regimes where fixed-scale density estimation leads to fragmentation and spurious modes. In this paper, we propose Doubly Stochastic Mean-Shift (DSMS), a novel extension that introduces randomness not only in the trajectory updates but also in the kernel bandwidth itself. We show that this randomized bandwidth policy can be beneficial for clustering and provide convergence theoretical results. Comparative experiments reveal that DSMS outperforms in many cases standard methods, exhibiting remarkable stability and preventing over-segmentation in sparse clustering scenarios without other performance degradation.

Keywords:
Unsupervised learning, Mean-shift clustering, Stochastic mean-shift, random kernel, non-parametric PDF estimation.

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1. Introduction

1.1. Motivation

Numerous applications of signal processing involve a clustering step, e.g. speaker diarization [1], image segmentation or superresolution [2, 3, 4]. In order to perform it, clustering algorithms have been proposed and investigated, among which the $k-$means [5], spectral clustering [6], DBSCAN [7], the well-known *Mean-shift* (MS) [8] and its variant the *blurring mean-shift* (BMS) [9]. Both MS and BMS algorithms can be considered as "deterministic" iterative procedures aiming to find local maximizers of an objective function, since they do not involve any random selection of points to perform their update rule. A general review of existing mean-shift algorithms can be found in [10], where the authors focus on acceleration schemes applied to the mean-shift framework, and provide a decision framework for selecting suitable acceleration strategies based on dataset characteristics and computational constraints.

In a recent publication [11], the authors introduced the Stochastic Mean-Shift (SMS), where the data sample points are selected randomly and moved by one iteration. It can still be considered to be a blurring process since the updated points are never replaced to their original locations and keep moving from their current position; at each step any sample point can be picked up for update. The algorithm proved to be convergent almost surely, as the gradient steps asymptotically tend to zero. This algorithm was successfully applied for speaker clustering tasks, where it outperformed MS in many cases. Though SMS represents an improvement of the ordinary MS, its properties remain to be investigated in depth. Indeed, stochastic algorithms can prove to be more efficient than deterministic methods for optimization

procedures [12].

One of the shortcomings of SMS- shared with other Mean-shift algorithms - lies in the fact that the kernel used has a fixed compact support. A fixed kernel size in most of mean-shift procedures introduces substantial limitations, as it is related to a data homogeneity assumption that does not hold in many setups. Fixed bandwidth kernels with radial symmetry also fail to account for anisotropic structures, where the relevant scale differs across directions and locations, thus preventing the algorithm from adapting to elongated or curved manifolds that frequently arise in practice [13]. It is therefore of great theoretical and practical interest to further investigate other strategies, which could significantly improve the stability and fidelity of mean-shift–based clustering and mode-seeking [14].

In order to remedy the aforementioned problem, this paper proposes to modify the SMS algorithm by introducing further a random scheme for the bandwidth, to be used for each position update of the selected point. Besides preventing convergence to false cluster centers partly, the hypothesis formulated is that random modifications of the kernel bandwidth in SMS allow data outliers to converge to actual modes, thus reducing the final number of obtained clusters. Randomization of fixed parameters, such as learning rates, has also shown to be beneficial to several learning algorithms, e.g. [15] and [16].

1.2. Related Works

Several techniques have been investigated in the past to improve MS's performance. Prior work has demonstrated that adaptive bandwidth strategies [10] can yield empirical improvements in convergence speed and clustering quality, by tailoring kernel weights to local density variations (typi-

cally via k-nearest neighbors), albeit at the cost of increased computational complexity [17]. In addition, [18] suggests mean shift updates with adaptive anisotropic kernel template, dynamically resized and reshaped to reduce tracking loss. This addresses a key limitation of traditional mean shift tracking, which uses a fixed or symmetric kernel and often fails when the target changes scale, shape, or appearance.

In other fields, recent k-means-based clustering methods have incorporated annealing to mitigate early convergence to poor local minima. Notable examples include Collaborative Annealing Power k-means++ [19] and Collaborative Annealing Fuzzy c-Means [20], where annealing is used to progressively sharpen cluster assignments and promote more reliable optimization trajectories.

While adaptive variants aim to optimize resolution at the individual point level by changing the scale, our method takes a different approach, using stochasticity as a way to improve the clustering process. Indeed, not only performing asynchronous "one-point-at-a-time" blurring updates (as in SMS), but it also plugs a randomly drawn bandwidth in the kernel weights. By this means, it mitigates the local mode traps encountered with fixed-scale estimators, and avoid binding each point to a fixed number of neighbors.

1.3. Paper overview

The improvement of SMS, called Doubly Stochastic Mean Shift (DSMS), is presented in section 2, theoretical results on the algorithm's convergence are presented and discussed in 3, and present results obtained on synthetic datasets in section 4. For a better comparison with SMS, we used the same synthetic datasets as in [11]. Proofs of the presented results, which do not rely on [11], are detailed in the appendix, whereas proofs resembling those

detailed in [11] are left to a supplementary material.

2. Doubly Stochastic mean-shift

The following section presents a stochastic version of MS based on random width kernel. Table 1 summarizes the notations used in this paper.

First, we provide details on the specific kernels to be considered in this work. These assumptions will be further used in the theoretical proofs.

Definition 1. *1. We call a profile function a mapping $k : [0,1] \to \mathbb{R}_+$ that is C^1, non-increasing, convex and such that $k'(t) < 0$ for all $t \in [0,1)$.*

2. a multivariate function which can be written as $K(\mathbf{u}) = k\left(\|\mathbf{u}\|^2\right)$ for all $\mathbf{u} \in \mathbb{R}^d$ will be referred to as a kernel with profile function k.

Typical examples of such profiles and associated kernels include $k(t) = (1-t)_+^\alpha$ for $\alpha = 2, 3, 4$, which lead respectively to the Bi-, Tri-, and Quad-weights kernels $K(\mathbf{u}) = \left(1 - \|\mathbf{u}\|^2\right)_+^\alpha$, respectively.

For any sample set $\{\mathbf{x}_i\}_{i\in[n]}$ with $\mathbf{x}_i \in \mathbb{R}^d, i = 1 \dots n$, stemming from some unknown multi-modal density f_X defined on $\mathbb{R}^d$, given a kernel K and a positive bandwidth parameter $h > 0$, the kernel density estimate (KDE) of f_X associated with $\{\mathbf{x}_i\}_{i\in[n]}$, denoted by D, verifies

$$\forall\, x \in \mathbb{R}^d \;\; D(x) \propto \frac{1}{nh^d} \sum_{i\in[n]} K\left(\frac{1}{h}\left(x - \mathbf{x}_i\right)\right). \tag{1}$$

The *mean-shift* algorithm [8, 21, 22] is a well known clustering method, based on finding the local maxima of (1). Since all the considered kernels have compact support by definition of the profile functions, this property is shared by any KDE defined in (1). For the rest of this paper, we shall

refer to the element $\mathcal{X} = [\mathbf{x}_1^\top, \ldots, \mathbf{x}_n^\top]^\top \in \mathbb{R}^{dn}$ as the *state* defined by this sample, and interpret any iterative clustering algorithm as a chain of transformations of $\mathbb{R}^{dn}$ applied to the initial state $\mathcal{X}^{(0)}$ given by the raw sample until we reach convergence (in a sense to be precised). A state shall be introduced further shortly as $\mathcal{X} = [\mathbf{x}_i]_{i\in[n]}$ for the sake of simplicity, without emphasizing further the implicit vector structure.

2.1. Blurring mean-shift and Stochastic mean-shift

For any state $\mathcal{X}$, we shall consider the *mean-shift operator* $\mathcal{S}_h : \mathbb{R}^d \to \mathbb{R}^d$ which maps a point $x \in \mathbb{R}^d$ to its new position as

$$\mathcal{S}_h(x;\mathcal{X}) = \frac{\displaystyle\sum_{i\in[n]} G\left(\frac{x-\mathbf{x}_i}{h}\right)\mathbf{x}_i}{\displaystyle\sum_{i\in[n]} G\left(\frac{x-\mathbf{x}_i}{h}\right)} \quad , \tag{2}$$

where $G(\mathbf{u}) = -k'\left(\|\mathbf{u}\|^2\right)$ defines the non-negative *weight function* associated with K. Note that the denominator in (2) is always positive since $-k'(0) > 0$. Note also that (2) is effectively computed on the $\mathcal{X}-$*neighborhood* of x which is defined as

$$N_h(x) = \left\{ i \in [n] \,|\, G\left(\frac{x-\mathbf{x}_i}{h}\right) \neq 0 \right\} = \left\{ i \in [n] \,|\, \|x-\mathbf{x}_i\| < h \right\},$$

Since K has a radial symmetry, the *shift vector* of x is defined as

$$\mathbf{m}_h(x;\mathcal{X}) = \mathcal{S}_h(x;\mathcal{X}) - x \quad ; \tag{3}$$

it can be easily verified that $\mathbf{m}_h(x)$ is indeed proportional to the gradient of D taken at x.

The Blurring mean-shift (BMS) is a kernel-based iterative method for data clustering. Starting from the initial state $\mathcal{X}^{(0)} \in \mathbb{R}^{dn}$ defined by the

original data sample, for any $k \in \mathbb{N}_0$ the kth step of the BMS consists in updating the current state $\mathcal{X}^{(k)}$ to the new state $\mathcal{X}^{(k+1)}$ by applying the mean-shift operator defined as follows:

$$\forall\, i \in [n] \quad \mathbf{x}_i^{(k+1)} = \frac{\sum_{j=1}^{n} G\left(\frac{\mathbf{x}_i^{(k)} - \mathbf{x}_j^{(k)}}{h}\right) \mathbf{x}_j^{(k)}}{\sum_{j=1}^{n} G\left(\frac{\mathbf{x}_i^{(k)} - \mathbf{x}_j^{(k)}}{h}\right)} \tag{4}$$

In other words, we apply to each sample point the mean-shift operator to find its new location, but by plugging in (2) the current state $\mathcal{X}^{(k)}$.

From a mathematical point of view, the BMS algorithm can be seen as an iterative procedure aimed at converging to a local maximizer of the following non-negative objective function

$$\forall\, \mathcal{X} = [\mathbf{x}_1^\top, \ldots, \mathbf{x}_n^\top]^\top \in \mathbb{R}^{d\,n} \quad L(\mathcal{X}) = \sum_{1 \leq i \leq j \leq n} K\left(\frac{1}{h}(\mathbf{x}_i - \mathbf{x}_j)\right) \quad ; \tag{5}$$

Indeed one can reformulate (4) as a "weighted" gradient step as follows:

$$\mathcal{X}^{(k+1)} = \mathcal{X}^{(k)} + \frac{h^2}{2} \sum_{i=1}^{n} \frac{1}{\sum_{j=1}^{n} G\left(\frac{\mathbf{x}_i^{(k)} - \mathbf{x}_j^{(k)}}{h}\right)} \nabla_{\mathbf{x}_i} L(\mathcal{X}^{(k)}) \tag{6}$$

where $\nabla_{\mathbf{x}_i} L(\mathcal{X})$ denotes the gradient component of L related to the $\mathbf{x}_i$ direction. Any state $\mathcal{X} \in \mathbb{R}^{dn}$ satisfies $\nabla L(\mathcal{X}) = \mathbf{0}_{dn}$ if and only if it is a fixed point of the transformation (4), i.e.

$$\forall\, i \in [n] \quad \mathbf{x}_i = \frac{\sum_{j=1}^{n} G\left(\frac{\mathbf{x}_i - \mathbf{x}_j}{h}\right) \mathbf{x}_j}{\sum_{j=1}^{n} G\left(\frac{\mathbf{x}_i - \mathbf{x}_j}{h}\right)} \tag{7}$$

The relation between the stationary points of L and our clustering objective is made explicit in a theorem from [23], which can be written as follows:

Theorem 1. *Assume the profile function k of the kernel K satisfies the assumptions of Definition 1. A state $\mathcal{X} = [\mathbf{x}_1^\top, \ldots, \mathbf{x}_n^\top]^\top \in \mathbb{R}^{dn}$ satisfies $\nabla L(\mathcal{X}) = \mathbf{0}_{dn}$ iff the following is satisfied:*

$$\forall\, (i,j) \in [n]^2 \;\; j \in N_h(\mathbf{x}_i) \iff \mathbf{x}_j = \mathbf{x}_i \tag{8}$$

In other words, for the specific truncated kernels considered in this study, a critical point of L is a state satisfying an ideal clustering condition: two points either coincide, or are distant from h at least. In the previous theorem, (8) will be later used to prove theoretical results on the effectiveness of the doubly stochastic mean-shift. Convergence of the BMS sequence has been thoroughly investigated and proved [24]; theorems establishing convergence rates for different kernel types can be found in [23].

The *stochastic mean-shift* algorithm (SMS) was presented in [11]. Roughly speaking, SMS is a blurring method like BMS but adds some randomness in the shifting process. Its principle is as follows: starting from an original state $\mathcal{X}^{(0)}$, in each subsequent step an index $i \in [n]$ is picked up randomly (according to the uniform distribution and independently of all previous choices), and the chosen point alone is moved by using the shift operator $\mathcal{S}_h$, by plugging in (2) the sample *current state* $\mathcal{X}^{(k)} = \left[\mathbf{x}_i^{(k)}\right]_{i=1}^n$ at step k. A summarized version of SMS is described in Algorithm 1a. Convergence diagnosis is either based on a maximal number of iterations or when $\|\mathbf{x}_i^{(k+1)} - \mathbf{x}_i^{(k)}\|$ is smaller than a user-defined threshold.

As noted in [11] SMS can be interpreted as some "partial" gradient algorithm: starting from some initial state $\mathcal{X}^{(0)} \in \mathbb{R}^{nd}$, the kth step of SMS consists in picking randomly an index $i_k \in \{1, \ldots, n\}$ and defining the $k+1$

state as

$$\mathcal{X}^{(k+1)} = \mathcal{X}^{(k)} + \frac{h^2/2}{\sum_{j=1}^{n} G\left(\frac{\mathbf{x}_{i_k}^{(k)} - \mathbf{x}_j^{(k)}}{h}\right)} \nabla_{\mathbf{x}_{i_k}} L(\mathcal{X}^{(k)}) \tag{9}$$

where $\nabla_{\mathbf{x}_i} L(\mathcal{X})$ denotes the partial gradient of L in the $\mathbf{x}_i$ direction. Algorithm 1a shows that SMS has a linear complexity $O(k)$ in the overall number of draws k, and computational advantages related to this property were discussed in [11].

2.2. Doubly stochastic mean-shift

In the SMS algorithm considered previously, the *bandwidth* (BW) h to be plugged in the kernel based update formula stays constant. In this work we introduce the *Doubly Stochastic mean-shift* (DSMS) which proceeds as the SMS, but also picks up randomly a bandwidth h_{k+1} at each step k (according to some specified distribution). The BW h_{k+1} at step k will be chosen inside a fixed positive interval $[h_{min}\,;\,h_{max}]$ for the following reason: too high values of h would likely expose the chosen point to the influence of too far points belonging to other clusters; and a too low value of h may lead to a poor update (if any). We shall define the BW *diameter* as $\Delta h = h_{max} - h_{min}$; experiments presented in section 4 will show empirically the influence of Δh on the method performance.

One version of the DSMS is presented in Algorithm 1b below, with associated block diagram in Figure 1; it shall be investigated in the theoretical section 3. This version is indeed one among others possible, since several ways to choose randomly h_k shall be compared in the experimental section 4. We shall assume some value $h_0 \in [h_{min}\,;\,h_{max}]$ is provided. Prior to the execution, some non-negative sequence (ν_k) converging to 0 is chosen.

Though its convergence rate is of no importance in the proofs, a slow rate is preferred in practice: in the experimental section we used $\nu_k \propto \frac{1}{\log(\log k)}$. Such a choice can be justified as follows: especially for samples with variable local densities, allowing h_k to cover a large domain of values proves beneficial to relate distant points to clusters and avoid clusters over-merging.

At each step k, given the BW h_{k-1} from the previous step, we compute first $\delta = \min\left\{\nu_k\,;\,\left(\frac{h_k}{h_{min}}\right)^2 - 1\,;\,1 - \left(\frac{h_k}{h_{max}}\right)^2\right\}$, then draw $\alpha = 1 + \delta\omega$ with $\omega \sim P_{[-1,1]}$, P being any probability distribution supported on $[-1, 1]$ having 0 mean. These choices ensure three things: first, the updated BW $h_{k+1} = \frac{h_k}{\sqrt{\alpha}}$ belongs to $[h_{min}\,;\,h_{max}]$; secondly, $h_{k+1} - h_k \to 0$ as k tends to infinity; last, it holds by definition of α that $\mathbb{E}[h_{k+1}^{-2}] = h_k^{-2}$, a point intervening in the proof of (13) in proposition 2. The last two conditions are used effectively in section 3 to prove the convergence of the algorithm. It should be precised that the sequences $(i_k), (h_k)$ are assumed to be independent.

Stated equivalently: after selecting randomly both an index $i_k \in [n]$ and a BW h_{k+1} the $k + 1^{\text{th}}$ state is defined as

$$\mathcal{X}^{(k+1)} = \mathcal{X}^{(k)} + \frac{h_{k+1}^2/2}{\sum_{j=1}^{n} G\left(\frac{\mathbf{x}_{i_k}^{(k)} - \mathbf{x}_j^{(k)}}{h_{k+1}}\right)} \nabla_{\mathbf{x}_{i_k}} L_{h_{k+1}}(\mathcal{X}^{(k)}), \tag{10}$$

where $\nabla_{\mathbf{x}_i} L_{h_{k+1}}(\mathcal{X})$ denotes the gradient of $L_{h_{k+1}}$ in the $\mathbf{x}_i$ direction.

3. Theoretical results

In this section we show that the DSMS provides almost surely a fixed clustering of the sample after a finite number of steps. Our theoretical analysis, both for the present paper and our previous work on SMS, draws in part on insights found in [23]. In particular, it relies on a similar non-decreasing property of the objective function proved in this reference (with

minor changes), as in SMS the bandwidth is kept constant throughout the whole process. Should the bandwidth becomes variable but non-decreasing, the objective values would be non-decreasing as well (as follows readily from the proposition 1 and the decreasing assumption on the profile function k). Such a scheme is undesired though, as it does not allow the bandwidth to oscillate between lower and upper values, which is quite valuable in a non-uniform local density context. More precisely, plugging increasingly higher bandwidth values may result in merging distinct clusters, whereas in low density regions higher bandwidth values may be useful to link those points to their respective clusters.

With a non-monotonic and randomly varying bandwidth, the sequence of objective values becomes random and non-monotonic (including asymptotically). As the convergence of the objective sequence and the convergence of the sample sequence are closely related (as shown in the aforementioned works), a natural way to tackle this problem is to find appropriate bandwidth schemes that guarantee almost-sure convergence of the objective. A common approach is to enforce classical assumptions from martingale theory, which is made possible given the convexity assumptions on the profile function. Proposition 2 below gives one particular type of random bandwidth scheme under which the objective sequence satisfies a submartingale property, without any claim to cover all possible schemes achieving (13).

Unless stated otherwise, in this section we suppose given any kernel K defined on $\mathbb{R}^d$, associated to a profile k satisfying all of the assumptions of Definition 1; all notations, definitions, and assumptions introduced there will be used in the present section. Unless stated otherwise, the results' proofs are postponed to the appendix.

3.1. Submartingale property of Doubly Stochastic Mean Shift

Starting from an initial state $\mathcal{X}^{(0)} \in \mathbb{R}^{dn}$, i.e. an array of n points $\mathcal{X}^{(0)} = [\mathbf{x}_1^\top, \ldots, \mathbf{x}_n^\top]^\top$ such that $\mathbf{x}_i \in \mathbb{R}^d$ for each i, we already introduced the typical DSMS k^{th} step in (10). It is immediate that the state $\mathcal{X}^{(k)}$ depends both on the random sequence of indices $[i_{k-1}, \ldots, i_0]$ and the random sequence of BW's $[h_k, \ldots, h_1]$, i.e. there exists a non-random function $F^{(k)} : ([n] \times [h_{min}\,;\, h_{max}])^k \to \mathbb{R}^{dn}$ such that $\mathcal{X}^{(k)} = F^{(k)}[(i_{k-1}, h_k); \ldots; (i_0, h_1)]$.

Fixing for a moment $h > 0$, and supposing we perform the kth update step in the SMS, the following proposition (proved in [11]) implies in particular that $L_h\left(\mathcal{X}^{(k+1)}\right) > L_h\left(\mathcal{X}^{(k)}\right)$ provided $\mathcal{X}^{(k+1)} \neq \mathcal{X}^{(k)}$:

Proposition 1. *Suppose the SMS with (fixed) BW h updates $\mathcal{X}^{(k)}$ to $\mathcal{X}^{(k+1)}$, it holds that*

$$L_h\left(\mathcal{X}^{(k+1)}\right) - L_h\left(\mathcal{X}^{(k)}\right) \geq \frac{2|\kappa'(0)|}{h^2} \|\mathbf{x}_{i_k}^{(k+1)} - \mathbf{x}_{i_k}^{(k)}\|^2 \tag{11}$$

with $i_k \in [n]$ denoting the randomly chosen index at step k. We have also the following upper bound on the gradient component along the direction $\mathbf{x}_{i_k}$:

$$\|\nabla_{i_k} L_h(\mathcal{X}^{(k)})\| \leq \frac{2\,n|\kappa'(0)|}{h^2} \left\|\mathbf{x}_{i_k}^{(k+1)} - \mathbf{x}_{i_k}^{(k)}\right\| \;; \tag{12}$$

This ascending property of the L_h values for a fixed h, combined to the particular way to draw randomly h_{k+1} given the value h_k which is exposed in algorithm 1b, has the interesting consequence that the sequence $\left(L_{h_k}(\mathcal{X}^{(k)})\right)$ is a discrete-time (positive) submartingale as we shall see now.

Proposition 2. *Let $0 < h_{min} < h_{max}$ be two positive constants. Given $h_0 \in [h_{min}, h_{max}]$, suppose that for any integer k we draw randomly $h_{k+1} = \dfrac{h_k}{\sqrt{1+\omega\delta}}$, for $\delta = \delta_k$ defined as in algorithm 1b, and with $\omega \sim P_{[-1,1]}$ being drawn according to a distribution P supported on $[-1, 1]$ and having zero*

mean i.e. $\int_{-1}^{1} \omega \, dP(\omega) = 0$. *Pick up randomly* $i_k \sim \mathcal{U}_{[n]}$ *and update the* i_k^{th} *point* $\mathbf{x}_{i_k}^{(k+1)} = \mathcal{S}_{h_{k+1}}(\mathbf{x}_{i_k}^{(k)}; \mathcal{X}^{(k)})$ *to define* $\mathcal{X}^{(k+1)}$. *It holds a.s. that*

$$\mathbb{E}\left[L_{h_{k+1}}(\mathcal{X}^{(k+1)}) \mid \mathcal{X}^{(k)}\,;\, h_k\right] \geq L_{h_k}(\mathcal{X}^{(k)}) \tag{13}$$

Thus the sequence $\left(L_{h_k}(\mathcal{X}^{(k)})\right)$ *is a discrete-time positive submartingale adapted to the process* $\left(\mathcal{X}^{(k)}\,;\, h_k\right)$.

Remark 1. *As can be understood from the previous proof, given* h_k *we could have drawn randomly* h_{k+1} *in many other ways than in algorithm 1b, and still satisfy the same submartingale property. We used here this particular scheme for the sake of proof simplicity.*

Since for any process sample ω one has $\forall\, k \in \mathbb{N}\ L_{h_k(\omega)}(\mathcal{X}^{(k)}(\omega)) \leq \frac{n(n+1)}{2}\,\kappa(0)$, we may apply a Doob's convergence theorem [25] to the positive submartingale $\left(L_{h_k}(\mathcal{X}^{(k)})\right)$: in our case there exists a bounded non negative function $L_\infty(\omega)$ such that $L_{h_k(\omega)}(\mathcal{X}^{(k)}(\omega)) \underset{k\to\infty}{\longrightarrow} L_\infty(\omega)$ almost surely on ω.

At each step k, (11) and (12) above imply the following

$$\|\nabla_{i_k} L_{h_{k+1}}(\mathcal{X}^{(k)})\|^2 \leq \frac{2\,n^2|\kappa'(0)|}{h_{k+1}^2}\left[L_{h_{k+1}}\left(\mathcal{X}^{(k+1)}\right) - L_{h_{k+1}}\left(\mathcal{X}^{(k)}\right)\right] \tag{14}$$

This has the following decisive consequence.

Corollary 1. *Keeping the same assumptions as in proposition 2, given any initial state* $\mathcal{X}^{(0)} \in \mathbb{R}^{dn}$*: for any DSMS sample process it holds a.s. that* $\nabla_{i_k} L_{h_{k+1}}(\mathcal{X}^{(k)}) \underset{k\to\infty}{\longrightarrow} \mathbf{0}$.

This result will in turn imply the existence a.s. of a fixed clustering after a finite number of steps.

3.2. Clustering and convergence results for DSMS

Let us introduce convenient notations for the rest of this section. At step $k \in \mathbb{N}$ of the DSMS algorithm, the gradient $\nabla L_h(\mathcal{X}^{(k)})$ for a BW value $h > 0$ shall be simply denoted by $\nabla_h^{(k)}$; hence $\nabla_{i\,;\,h}^{(k)}$ shall denote $\nabla_{\mathbf{x}_i} L_h(\mathcal{X}^{(k)})$ for any $i \in [n]$; and $\|\nabla_h^{(k)}\|$ shall denote the norm $\max_{i\in[n]} \|\nabla_{i\,;\,h}^{(k)}\|_2$ i.e the maximal component wise ℓ_2 norm value of the gradient $\nabla_h^{(k)}$.

In this subsection we shall consider the process $\left(\mathcal{X}^{(k)}\,;\,h_k\right)$ which takes its values in the space $\mathbb{R}^{nd} \times [h_{min}\,;\,h_{max}]$. It is by our assumptions a non homogeneous Markov chain [26]. For any $k \in \mathbb{N}$ we denote by α_k the random coefficient such that $h_{k+1} = h_k/\sqrt{\alpha_k}$, so that $h_{k+1} > h_k$ if $\alpha_k < 1$. We introduce now definitions and notations that will be used in subsequent proofs. Suppose given any $\epsilon > 0$ and fixed positive integer k_0 ; we define the event $\mathcal{A}_{k_0} = \left\{\omega \mid \forall\, k \geq k_0 \ \|\nabla_{i_k\,;\,h_{k+1}}^{(k)}(\omega)\|_2 < \epsilon\right\}$. By previous considerations from subsection 3.1, every process sample belongs a.s. to an $\mathcal{A}_{k_0}$ for some $k_0 \in \mathbb{N}$.

For a given $\epsilon > 0$ as before, let us also introduce a sequence of stopping times $(T_p)_{p\in\mathbb{N}}$ for the stochastic process $\left(\mathcal{X}^{(k)}\,;\,h_k\right)$ which shall be used in the proofs. Let us define $T_1 = \min\left\{k \in \mathbb{N} \mid \|\nabla_{h_{k+1}}^{(k)}\| \geq \epsilon\right\}$ (we set $T_1 = \infty$ is there is no such k); for any $p \geq 2$ we define $T_p = \min\left\{k > T_{p-1} \mid \|\nabla_{h_{k+1}}^{(k)}\| \geq \epsilon\right\}$ (set to ∞ if there is no such integer k). The sequence $(T_p)_{p\in\mathbb{N}}$ is clearly adapted to the sequence $[(i_k\,;\,h_{k+1})]_{k\in\mathbb{N}}$ which determines the process sample.

Lemma 1. *For any $\epsilon > 0$, $k_0 \in \mathbb{N}$ and $p \in \mathbb{N}_0$ we have*

$$\mathbb{P}\left[(T_{k_0+p} < \infty)\,\mathcal{A}_{k_0}\right] \leq \left(1 - \frac{1}{n}\right)^p \tag{15}$$

The previous lemma has the following important consequence:

Proposition 3. *Given any initial state $\mathcal{X}^{(0)} \in \mathbb{R}^{dn}$, and under the same assumptions as in proposition 2, the DSMS sequence satisfies a.s. that*

$$\|\nabla^{(k)}_{h_{k+1}}\| \underset{k\to\infty}{\longrightarrow} \mathbf{0}_{dn}$$

The main clustering result is now presented. It states that almost surely, after a specific step in the SMS, well-separated and stable clusters are observed.

Theorem 2. *Under the same assumptions as in proposition 2, given any initial state $\mathcal{X}^{(0)} \in \mathbb{R}^{dn}$, let $\left(\mathcal{X}^{(k)}\right)$ be the DSMS process sample. Suppose τ is any given positive real s.t. $\tau < \frac{h_{min}}{2}$. Then there exists a.s some partition $J_1, \ldots, J_M$ of $[n]$ and $K \in \mathbb{N}$ such that for every $k \geq K$:*

1) $\forall\, l \in [M]$ *it holds that* $\max\limits_{\substack{i\in J_l \\ j\in J_l}} \|\mathbf{x}_i^{(k)} - \mathbf{x}_j^{(k)}\| < \tau$;

2) $\forall\, (l,m) \in [M]^2$ *s.t* $l \neq m$, *it holds that* $\min\limits_{\substack{i\in J_l \\ j\in J_m}} \|\mathbf{x}_i^{(k)} - \mathbf{x}_j^{(k)}\| > h_{min} - \tau$;

in other words a.s. the points $\mathbf{x}_1^{(k)}, \ldots, \mathbf{x}_n^{(k)}$ belong to fixed clusters for sufficiently large k, the lth cluster being defined as $\mathcal{C}_l = \left\{\mathbf{x}_i^{(k)}\,,\, i \in J_l\right\}$; moreover all clusters diameters tend a.s to 0 as $k \to \infty$.

For the sake of completeness, proofs of lemma 1 and theorem 2 above are provided in the supplementary material section. Both follow closely similar proofs from [11].

3.3. Practical Convergence Diagnosis

Given any initial state, the minimal number k of DSMS steps which are necessary to achieve the aforementioned clustering is by nature a random variable. No theoretical results in this paper provide further precision about

a higher bound on k which would hold under a specified high probability. By the one-shifted-point policy, each SMS step costs linearly in n [11], a property shared by DSMS. As a result, both algorithms might need more steps than their BMS or MS counterparts to achieve the desired clustering. This constraint makes it necessary to define convenient stopping criteria, even if the clustering above has not been completely achieved yet. In practice, whenever a sufficient number of points have a sufficiently small last shift, the algorithm stops. Such a practice is justified by the conclusions of propositions 1 and 3.

4. Numerical Experiments

We present in this section the results obtained on synthetic data. The distribution of the samples and the clusters used were similar to [11], for the sake of a fair comparison with the other algorithm. DSMS was compared with MS, BMS and SMS.

For quantitative examinations, we used the average number of clusters, the Average Cluster Purity (ACP) and the Average Label Purity (ALP), both defined as in [1, 27, 28]:

$$ACP = \frac{1}{Q} \sum_{q=1}^{Q} \sum_{r=1}^{R} P(d_r|c_q)^2 \tag{16}$$

$$ALP = \frac{1}{R} \sum_{r=1}^{R} \sum_{q=1}^{Q} P(c_q|d_r)^2 \tag{17}$$

where in the latter $P(d_r|c_q)$ is the probability for a point of the cluster c_q to have the true label d_r, and $P(c_q|d_r)$ is similarly defined. The inner sums in (16) and (17) defines a purity index for each individual cluster and label respectively. Roughly speaking, a low ACP indicates that the clusters

are not class homogeneous, whereas low ALP points to classes decomposed along several clusters. Taking both aspects into account can be done by considering the geometric average:

$$K = \sqrt{ACP \cdot ALP} \quad \in (0, 1] \tag{18}$$

4.1. Performance on Underrepresented Clusters

In numerous applications, some clusters are not well represented. For example, in speaker diarization, clustering is harder for speakers intervening casually in the recorded discussion. In forensic speech processing, and specifically for short-duration speaker diarization, the system is required to distinguish between multiple speakers based on very few speech segments [29]. In these regimes, traditional algorithms are plagued by noise overfitting and spurious local modes, requiring robust stochastic approaches that can infer stable clusters despite relying on little information.

In order to investigate the performance of DSMS in this framework, the following experiments were performed. The data consists of a 2-dimensional Gaussian Mixture Model (GMM) with 3 isotropic clusters, whose parameters are summarized in Table 2. The number of points per cluster is variable to test the algorithm's performance on sparse data. Though this model might appear simplistic at first glance, it allows us to clearly identify the influence of the hyperparameters for each algorithm and to point out their most critical shortcomings. The number of sample points was made equal for each cluster, and ranged from 10 to 200, in order to investigate the number of found clusters after convergence. Concerning the random scheme used to draw a bandwidth at each step, we tested two types of continuous distributions for P (see algorithm 1b), namely uniform and beta (with no notable differences in the results). The hyperparameters for each algorithm

are summarized in Table 3. For each setup, the experiments were performed 100 times. Figure 2 illustrates the average number of clusters found using MS, BMS, SMS, and DSMS, with their associated 90% confidence interval. We observe that DSMS largely outperforms SMS and MS, and estimates the clusters number better than BMS. These improvements on underrepresented clusters are explained by the random bandwidth grouping together sample points that other algorithms would consider outliers and leave unchanged.

While the gap between the investigated methods is less significant when the sample size is sufficiently large ($N > 100$), a distinct difference emerges in the sparse regime ($10 \leq N \leq 50$). Indeed, MS and BMS exhibit a tendency toward *over-segmentation* as the number of points per cluster decreases. This phenomenon is consistent with the limitations of fixed-bandwidth kernel density estimation, as scarce data can be interpreted by a fixed bandwidth procedure as multiple modes, thereby increasing artificially the number of estimated clusters. SMS, while more robust to local noise due to its asynchronous update rule, eventually suffers from the same limitation. The figure indicates that without scale flexibility, stochasticity alone cannot bridge the significant density gaps present in small datasets. In contrast, by randomly varying both the selected point and the kernel size at each iteration, DSMS achieves improved performance by leveraging multi-scale neighborhood information, maintaining a stable estimate of the cluster count across the entire range of sample sizes.

While smaller bandwidth samples refine mode localization, larger bandwidths prove beneficial for samples in low-density regions. Importantly, next results will show that this improvement does not degrade the clustering performance in terms of K. These observations also indicate the need to take the data repartition into account in the bandwidth sampling strategy, in

order to optimize the clustering process. Data-dependent sampling strategies are out of the scope of the present paper and should be investigated in future contributions.

4.2. Runtime Analysis

Unsurprisingly, stochastic variants of MS exhibit higher runtimes due to their slower convergence dynamics. This is illustrated in Table 7, which details the average runtime over 100 runs of the Mean-shift-based approaches across all tested datasets, with a maximal number of iterations equal to 5.10^5. Though these large computation times appear problematic for real-time applications, it should be noted that in our experiments, the maximal number of iterations was set large voluntarily to get a good convergence to the clusters' centers. However, satisfactory clustering capabilities can already be obtained with 10 times fewer iterations, as shown by Figure 9. This figure illustrates the need to investigate in future contributions good convergence diagnosis strategies, as well as acceleration techniques, e.g. by using additional information on the data's local.

Unlike classical MS, the trajectory of DSMS constitutes a discrete-time Markov chain governed by stochastic transition kernels. Consequently, diagnosing or deriving an explicit analytical bound on the algorithm's rate of convergence is beyond the scope of this work. To address this from a practical standpoint, we provide an empirical execution time analysis in Fig. 3, plotting runtime against the number of samples n on a log-log scale for the experiment on synthetic Gaussian data. Results suggest that the empirical stopping time obeys a stable polynomial rate, ensuring that the stochastic execution framework, though less appealing in terms of speed than MS or BMS, remains computationally tractable in large-scale data regimes.

4.3. Exhaustive Comparison with SMS

The second experiment investigates any potential loss of performance when moving from SMS to DSMS. For this purpose, we uses the same experiments as described in [11] (we refer to the same publication for an in-depth description of the experimental setup and parameters' choice). These experiments evaluate performance of both SMS and DSMS as a function of the number of clusters and the class imbalance. In particular, Figures 4 and 5 show the evolution of K for both SMS and DSMS as a function of the ratio between clusters cardinals and the number of clusters, respectively. All results presented in subsection 4.5 with their 90%-confidence interval, seem to demonstrate that provided the BW diameter $\Delta h = h_{max} - h_{min}$ is conveniently chosen, no loss of performance is observed. Therefore, in such a case, DSMS should be preferred over its SMS counterpart for better clustering results, as detailed previously.

4.4. Comparison with other methods on common datasets

In this subsection, we compare extensively DSMS to all of the following methods: MS, BMS and SMS, but also DBSCAN [7, 30], K-means [5], and Spectral Clustering [31]. These algorithms were evaluated on both synthetic and real low-dimensional datasets, as well as on the high-dimensional Isolet speech-recognition dataset. To make comparisons as fair as possible, we searched for MS, DBSCAN and Spectral Clustering optimal hyperparameter values in terms of K and number of classes. In contrast, DSMS does not rely on such prior information.

The first evaluation round used synthetic datasets Asets and Shapes (Aggregation, Compound, Jain, Spiral) [32], and the real datasets Iris, Seeds, Banknote, and Behavior. For reproducibility, optimal hyperparameter val-

ues used in these cases are presented in table 4. Table 5 displays the results (averaged over 100 runs of each algorithm) of all the methods investigated. An example of clustering is shown in Figure 7. Overall, DSMS achieves the best performance in most cases among the Mean-Shift variants (MS, BMS, SMS). On datasets like Aggregation, Iris and Behavior, DSMS provides the most accurate estimation of N and the highest K scores. This confirms that the doubly stochastic approach successfully prevents the algorithm from getting trapped in fake modes. The performance on Iris and Behavior is particularly noteworthy. DSMS achieves the highest overall metrics here, suggesting a stochastic bandwidth can be beneficial for real-world data. Despite its strengths, the data highlights three specific areas where DSMS faces challenges. The most obvious limitation is seen in the Spiral dataset. While DSMS ($K = 0.37$) is a slight improvement over other MS variants, it is utterly outperformed by Spectral Clustering ($K = 1.0$) and DBSCAN ($K = 0.99$). Also, in the Compound dataset, DSMS ($K = 0.84$) performs well but lags behind DBSCAN (when the epsilon is tuned correctly), and spectral clustering. These limitations are commented below at the end of this subsection.

Also, DSMS appears slightly more robust to noise and varying density than DBSCAN on Iris and Behavior, but DBSCAN is superior when the "shape" of the cluster prevails over its "center" (e.g., Banknote). Spectral Clustering remains the best choice for complex geometries (Spiral, Jain), but it requires a pre-defined number of clusters. DSMS offers a valuable "middle ground"— handling geometry much better than k-means or MS, wit no prior knowledge of the clusters number.

In an additional real data evaluation, all methods were tested on the high-dimensional UCI speech dataset Isolet, commonly used for isolated let-

ter recognition (617 dimenional with 26 ground-truth classes). The results, presented in table 6, show that DSMS and BMS offer the best performances for classes number estimate among non-parametric methods (with 61 and 55 clusters found respectively on average). Also, both perform similarly in terms of the K measure, slightly better than their MS and SMS counterparts.

While DSMS demonstrates good capabilities in numerous cases, its limitations are common to mean-shift based methods. On complex topological structures such as Spiral and Compound (where DBscan is better), and also Seeds (where Spectral Clustering rules), DSMS exhibits poorer performances: indeed, Mean-Shift algorithms, including DSMS, typically rely on a multimodal assumption for the local density. However, manifolds such as a spiral may exhibit no well-isolated high-density modes, which leads to over-clustering (See also the supplementary material for additional visualization). Another limitation emerges in environments featuring simultaneously varying densities and multi-scale structures, such as Compound. Although DSMS performs reasonably well, it lags behind simpler methods like DBSCAN. While the stochastic bandwidth sampling of DSMS aids in overcoming local noise, it remains bounded by the choice of the bandwidth distribution.

4.5. Influence of the bandwidth range

In this subsection we discuss specifically the crucial issue of bandwidth scheme specifications for DSMS. Algorithm 1b gives one possible way to draw randomly the bandwidth sequences, both effective in the experiments and ensuring a stable clustering is reached almost surely after a finite number of steps. The global range to which the bandwidths belong is of great

interest: indeed the slow ν_k convergence rate to 0 in table 3 ensures that as k tends to infinity, the bandwidths sequence (h_k) will cover a significant range of values between two user-defined bounds h_{min} and h_{max}. In particular, and consistently with what we mentioned in subsection 4.3, a key factor distinguishing SMS from DSMS in terms of performance lies in the choice of the range $[h_{min}\,;\,h_{max}]$, and particularly of the BW diameter Δh. Such a choice is essentially a trade-off matter, as drawing larger h_k's facilitates the merging of fragmented components but risks merging distinct clusters, whereas in contrast drawing lower h_k's restricts the ability to bridge low-density regions, potentially leading to over-segmentation. Also, too low a Δh hardly makes any difference with the SMS.

We especially evaluated the influence of Δh on 100 runs of DSMS on 3 clusters with distribution similar as in Table 2. Each clusters included 100 data samples, and each run kept the data identical for all the changes in bandwidth range to avoid random artifacts. For each experiment, the limit bandwidth $h_\infty \in [h_{min}, h_{max}]$ was chosen to get optimal results for SMS. Figure 6 presents the evolution of the parameter K as Δh increases.

Notably, experiments indicate the existence of a domain of bandwidth diameters allowing better clustering than SMS, whereas performance drops significantly for higher Δh values. This behavior can be explained by the analysis of the ACP and ALP, presented in Figure 8. Indeed, for a reasonably high Δh, the ALP score (a proxy for label recall) improves by the increased ability to merge fragments of the same class, and integrate outliers. But as Δh becomes higher, some clusters are corrupted with points from neighboring classes, degrading the ACP score. We provide in the supplementary material a visual inspection of the effect of bandwidth on the Aggregation parameter. Besides the bandwidth parameters h_{min}, h_{max} and

h_∞, the parameter ν_k influences the speed of convergence and the possibility of improving clustering capabilities even if the limit radius is not carefully chosen. We illustrate this aspect numerically in Table 8 on the Aggregation dataset, and provide graphical examples and a more in-depth discussion of this aspect in the supplementary material.

The crossing point of these opposing trends explains the stability region observed in the evolution of K. These results suggest that for specific topologies, introducing a reasonable amount of stochasticity in the choice of bandwidths yields a better balance between class separation on the one hand, and robustness to intra-class variance on the other. Generally speaking, selecting a proper bandwidth at a given point is typically a local issue, meaning in our case that we might update the bandwidth randomly, but still in accordance with the local landscape around this point. Such additional insights may yield an improvement for our method, and shall be addressed in future work.

5. Conclusion

We presented DSMS, an improvement of the Stochastic Mean Shift algorithm where the radii around data samples are drawn randomly. Theoretical proofs including the existence a.s. of a stable clustering after a finite number of steps were presented, and results on synthetic dataset showed that this novelty enables a more accurate identification of sparse clusters, without any degradation in its performance on the remaining criteria relative to SMS.

On numerous low and high-dimensional datasets, DSMS frequently outperformed other state-of-the-art methods, illustrating its robustness and suitability for a variety of conditions.

This work also indicates that intrinsic structure is rarely confined to a single scale, and that by integrating a random bandwidth policy, a known clustering process can be made more resilient to data scarcity and outliers. Further works will emphasize these aspects, and develop suitable sampling strategies towards optimality.

Appendix A. Proof of Proposition 1

The proof of the first inequality can be found in see [11]. The second assertion is an immediate consequence of the fact that $\nabla_{i_k} L(\mathcal{X}^{(k)}) = \frac{2}{h^2} \sum_{j=1}^{n} G\left(\frac{\mathbf{x}_{i_k}^{(k)} - \mathbf{x}_j^{(k)}}{h}\right) \left[\mathcal{X}^{(k+1)} - \mathcal{X}^{(k)}\right]$, taking into account that $G(\boldsymbol{u}) = |\kappa'(\|\boldsymbol{u}\|^2)|$ is a non increasing function of the radius $\|\boldsymbol{u}\|$.

Appendix B. Proof of Proposition 2

We start with the following inequality implied by proposition 1:

$$\begin{aligned} L_{h_{k+1}}(\mathcal{X}^{(k+1)}) - L_{h_k}(\mathcal{X}^{(k)}) &= L_{h_{k+1}}(\mathcal{X}^{(k+1)}) - L_{h_{k+1}}(\mathcal{X}^{(k)}) \, + \, L_{h_{k+1}}(\mathcal{X}^{(k)}) - L_{h_k}(\mathcal{X}^{(k)}) \\ &\geq L_{h_{k+1}}(\mathcal{X}^{(k)}) - L_{h_k}(\mathcal{X}^{(k)}) \end{aligned}$$

We shall prove that $\mathbb{E}\left[L_{h_{k+1}}(\mathcal{X}^{(k)}) \mid \mathcal{X}^{(k)} \, ; \, h_k\right] \geq L_{h_k}(\mathcal{X}^{(k)})$. Let us denote for simplicity $d_{i,j} = \|\mathbf{x}_i^{(k)} - \mathbf{x}_j^{(k)}\|$ for all $(i,j) \in [n]^2$, then by definition $L_{h_{k+1}}(\mathcal{X}^{(k)}) = \sum_{1 \leq i \leq j \leq n} \kappa\left(d_{i,j}^2 / h_{k+1}^2\right)$, hence using the definition of h_{k+1} and Jensen's inequality by the convexity assumption on the profile function κ we obtain the following:

$$\mathbb{E}\left[L_{h_{k+1}}(\mathcal{X}^{(k)}) \mid \mathcal{X}^{(k)} \, ; \, h_k\right] = \sum_{1 \leq i \leq j \leq n} \int_{(1-\delta \, ; \, 1+\delta)} \kappa\left(\alpha \frac{d_{i,j}^2}{h_k^2}\right) dP(\alpha) \qquad \text{(B.1)}$$

$$\geq \sum_{1 \leq i \leq j \leq n} \kappa\left(\frac{d_{i,j}^2}{h_k^2}\right) = L_{h_k}(\mathcal{X}^{(k)}) \qquad \text{(B.2)}$$

since by our choice of the distribution P of α on $[1-\delta, 1+\delta]$ we have $\int_{1-\delta}^{1+\delta} \alpha \, dP(\alpha) = 1$. The proposition is proved.

Appendix C. Proof of Corollary 1

Indeed (14) above implies that for any $k \in \mathbb{N}$:

$$\|\nabla_{i_k} L_{h_{k+1}}(\mathcal{X}^{(k)})\|^2 \leq \frac{2\,|\kappa'(0)|\,n^2}{h_{min}^2} \left[L_{h_{k+1}}\left(\mathcal{X}^{(k+1)}\right) - L_{h_k}\left(\mathcal{X}^{(k)}\right) + \left(L_{h_k} - L_{h_{k+1}}\right)(\mathcal{X}^{(k)}) \right] \tag{C.1}$$

First it holds a.s. that $L_{h_{k+1}}\left(\mathcal{X}^{(k+1)}\right) - L_{h_k}\left(\mathcal{X}^{(k)}\right) \underset{k\to\infty}{\longrightarrow} 0$ by submartingale convergence. At any step k the updated point belongs to the convex envelope $\mathrm{Conv}\left(\mathcal{X}^{(0)}\right)$ in $\mathbb{R}^d$ of the points composing $\mathcal{X}^{(0)}$: indeed (10) shows that $\mathrm{Conv}\left(\mathcal{X}^{(k+1)}\right) \subseteq \mathrm{Conv}\left(\mathcal{X}^{(k)}\right)$, and the claim follows by straightforward induction. By uniform continuity of the map $(\mathcal{X}, h) \mapsto L_h(\mathcal{X})$ on the compact $\mathrm{Conv}\left(\mathcal{X}^{(0)}\right)^n \times [h_{min}, h_{max}]$, and the condition $h_{k+1} - h_k \to 0$ it follows that $\left(L_{h_k} - L_{h_{k+1}}\right)(\mathcal{X}^{(k)}) \to 0$ does hold. The corollary is proved.

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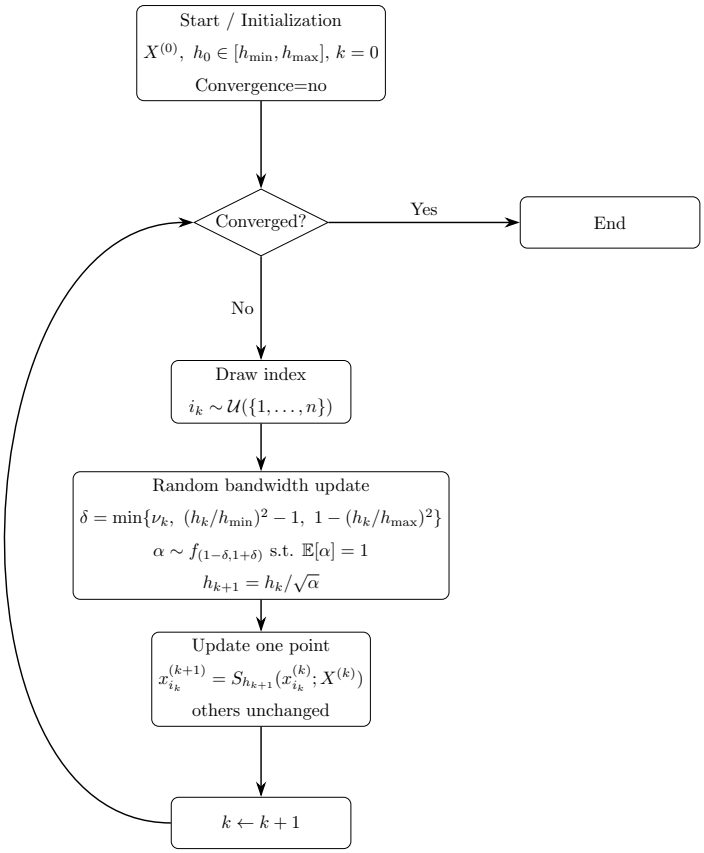


Figure 1: Block diagram of DSMS.

Table 1: Summary of the main notations used throughout the paper

Notation	Significance
General notations	
$[n]$	the set of integers $\{1, \dots, n\}$
$\mathbb{R}_+$	the set of non negative reals $[0, \infty)$
$\|\mathbf{u}\|\,,\ \mathbf{u} \in \mathbb{R}^{n\times 1}$	the ℓ_2 norm of $\mathbf{u}$
Specific notations	
$x_1, \dots x_J$	The J data points
$K(\mathbf{u}) = k\left(\|\mathbf{u}\|^2\right)\ ;\ G(\mathbf{u}) = -k'\left(\|\mathbf{u}\|^2\right)$	density kernel K on $\mathbb{R}^n$ associated to the profile $k : [0,\infty) \to \mathbb{R}_+$; "weight" function on $\mathbb{R}^n$
$N_h(x),\ x \in \mathbb{R}^d$	subset of $[n]$ of the neighboring indices $\{i : \|x - \mathbf{x}_i\| < h\}$
$\mathcal{X} = [\mathbf{x}_i]_{i\in[n]} \in \mathbb{R}^{nd}$	a state, i.e. a column vector made of the ordered collection of n points each one in $\mathbb{R}^d$
$L_h(\mathcal{X})$	the cost function $\sum_{1\le i\le j\le n} K\left(\frac{\mathbf{x}_i - \mathbf{x}_j}{h}\right)$ at state $\mathcal{X}$, depending on the bandwidth h
i_k	the index $i \in [n]$ randomly chosen at the kth step of DSMS
h_k	the BW at the kth step of DSMS
$\nabla_i L_h$	partial gradient $\nabla_{\mathbf{x}_i} L_h$ of L_h w.r.t the variable $\mathbf{x}_i$
$\nabla_h^{(k)}\ ;\ \nabla_{i;h}^{(k)}$	gradient of L_h at step k of DSMS ; partial gradient $\nabla_i L_h$ at step k of DSMS
$\|\mathbf{U}\|\,,\ \mathbf{U} = [\mathbf{u}_1^T, \dots, \mathbf{u}_n^T]^T\,,\ \mathbf{u}_i \in \mathbb{R}^{d\times 1}$	$\max_{i\in[n]} \|\mathbf{u}_i\|_2$: maximal component wise ℓ_2 norm value
$[h_{min}, h_{max}]$	the interval of considered bandwidths
$\Delta h := h_{max} - h_{min}$	BW diameter i.e. the length of $[h_{min}\,;\, h_{max}]$

Table 2: Data generation hyperparameters for the 3-Gaussian Mixture Model. The values for Mean, Standard Deviation, and Sample Size are detailed for each Cluster ID.

	Cluster ID		
Parameters	**Cluster 1**	**Cluster 2**	**Cluster 3**
Mean Vector	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -1 \\ -1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
Covariance matrix	$0.65\mathbf{I}_2$	$0.65\mathbf{I}_2$	$0.65\mathbf{I}_2$
Points per Cluster (N)	$N \in [10; 200]$	$N \in [10; 200]$	$N \in [10; 200]$

while Convergence is not attained **do**

Draw the index $i_k \in [n]$ uniformly

$\mathbf{x}_{i_k}^{(k+1)} \leftarrow \mathcal{S}_h(\mathbf{x}_{i_k}^{(k)}; \mathcal{X}^{(k)})$

$k + 1 \leftarrow k$

Convergence diagnosis

end while

(a) Stochastic Mean-Shift

choose any sequence (ν_k) s.t. $\nu_k \to 0$

fix any probability distribution P on $[-1, 1]$ s.t. $\int_{-1}^{1} \omega \, dP(\omega) = 0$

while Convergence is not attained **do**

Draw the index $i_k \in [n]$ uniformly

Define $\delta = \min\left\{\nu_k \,;\, \left(\frac{h_k}{h_{min}}\right)^2 - 1 \,;\, 1 - \left(\frac{h_k}{h_{max}}\right)^2\right\}$

Draw $\omega \sim P$ in $[-1, 1]$ and define $h_{k+1} = \frac{h_k}{\sqrt{1+\delta\omega}}$

$\mathbf{x}_{i_k}^{(k+1)} \leftarrow \mathcal{S}_{h_{k+1}}(\mathbf{x}_{i_k}^{(k)}; \mathcal{X}^{(k)})$

$k + 1 \leftarrow k$

Convergence diagnosis

end while

(b) Doubly Stochastic Mean-Shift

Algorithms 1: Stochastic and Doubly Stochastic mean-shifts.

Table 3: Hyperparameter settings for Mean-Shift (MS), Blurring Mean-Shift (BMS), Stochastic Mean-Shift (SMS), and Doubly Stochastic Mean-Shift (DSMS).

Parameter	**MS**	**BMS**	**SMS**	**DSMS**
	Common Parameters			
Max Iterations	10^7			
Convergence Threshold	10^{-6}			
	Algorithm-Specific Parameters			
Bandwidth (h)	0.6	0.6	0.6	*Variable*
Bandwidth Range $[h_{min}, h_{max}]$	N/A	N/A	N/A	$[0.2, 1.6]$
Limiting Bandwidth h_{inf}	N/A	N/A	N/A	0.6
ν_k Convergence Rate	N/A	N/A	N/A	$\frac{1}{\log_{10}(10 + \log_{10} k)}$

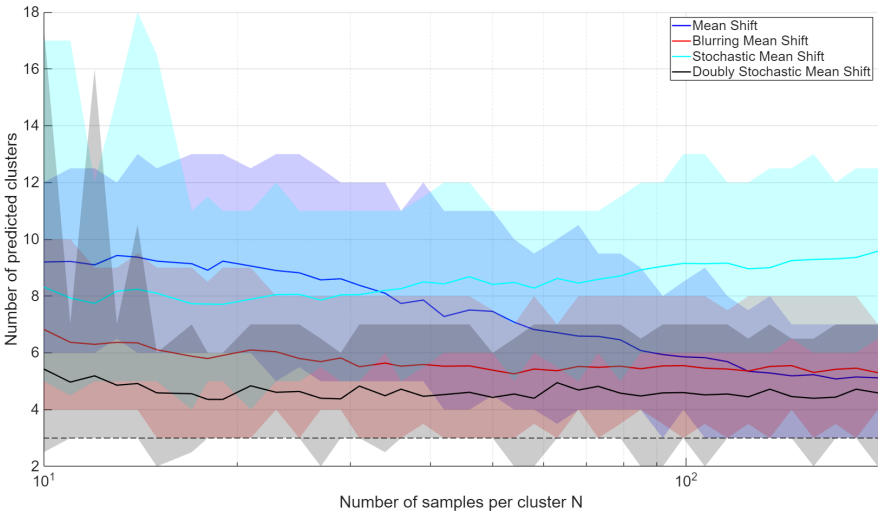


Figure 2: Average number of clusters with 90% confidence intervals for the different algorithms, as a function of the number of samples per cluster. The true number of clusters (3) is represented by the thin dotted line.

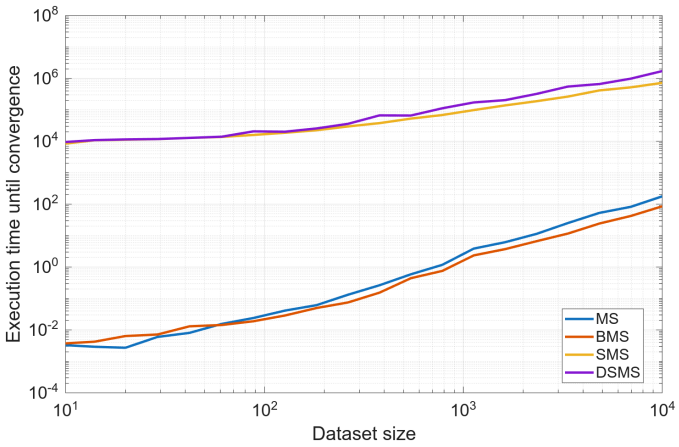


Figure 3: Execution times as a function of the number of samples in the synthetic Gaussian experiments.

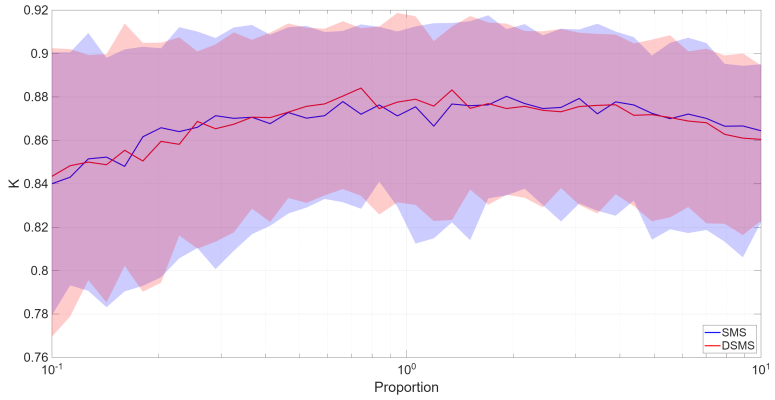


Figure 4: K as a function of the class imbalance ratio as defined in [11]

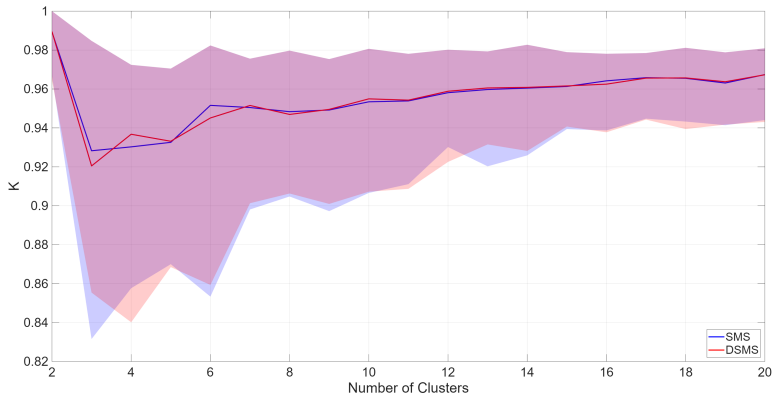


Figure 5: K as a function of the number of clusters

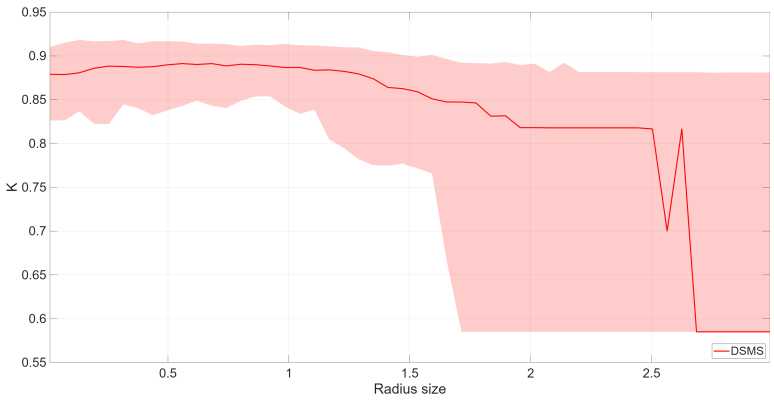


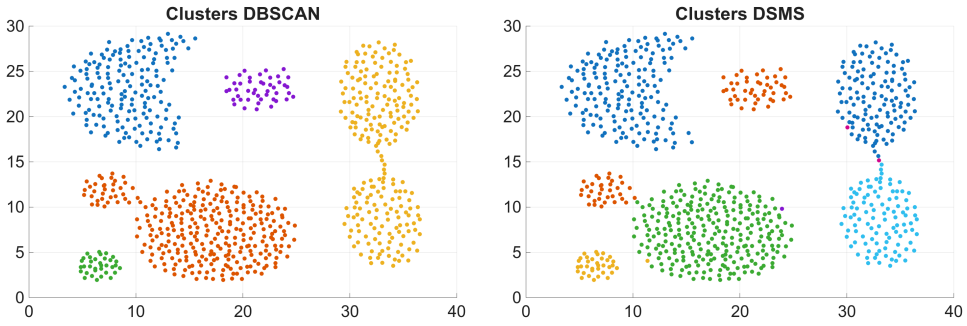
Figure 6: K as a function of the BW diameter $\Delta h = h_{max} - h_{min}$

Dataset	**Hyperparameters**				
	MS,BMS, SMS	DSMS	DBSCAN	K-means	Spectral Clustering
Aggregation	$h = 2.5$	$h_{min} = 1$, $h_{max} = 4$	$\epsilon = 1.4$, MinPts $= 5$	$k = 7$	$k = 7$ $\sigma = 2$
Compound	$h = 2.15$	$h_{min} = 0.8$, $h_{max} = 3.5$	$\epsilon = 1.3$, MinPts $= 5$	$k = 6$	$k = 6$ $\sigma = 1.5$
Spiral	$h = 2$	$h_{min} = 0.8$, $h_{max} = 3.2$	$\epsilon = 2$, MinPts $= 5$	$k = 3$	$k = 3$ $\sigma = 1.5$
Jain	$h = 4.5$	$h_{min} = 3.5$, $h_{max} = 6$	$\epsilon = 2.5$, MinPts $= 5$	$k = 2$	$k = 2$ $\sigma = 2$
A3	$h = 3000$	$h_{min} = 1000$, $h_{max} = 4000$	$\epsilon = 775$, MinPts $= 5$	$k = 21$	$k = 21$ $\sigma = 4000$
Iris	$h = 0.859$	$h_{min} = 0.8$, $h_{max} = 1$	$\epsilon = 0.5$, MinPts $= 5$	$k = 3$	$k = 3$ $\sigma = 0.6$
Seeds	$h = 2.5$	$h_{min} = 2$, $h_{max} = 3$	$\epsilon = 0.59$, MinPts $= 10$	$k = 3$	$k = 3$ $\sigma = 0.54$
Banknote	$h = 5.5$	$h_{min} = 4$, $h_{max} = 7$	$\epsilon = 2.5$, MinPts $= 9$	$k = 2$	$k = 2$ $\sigma = 2.5$
Behavior	$h = 0.35$	$h_{min} = 0.3$, $h_{max} = 0.45$	$\epsilon = 0.23$, MinPts $= 9$	$k = 4$	$k = 4$ $\sigma = 0.1$

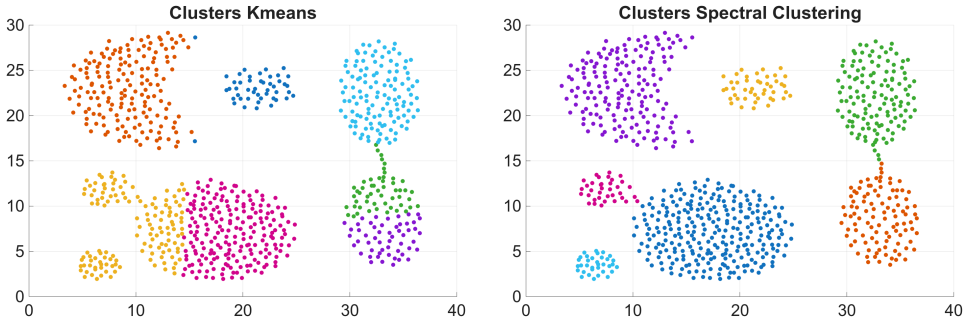
Table 4: Hyperparameters used for the comparison of algorithms on several datasets. Parameters h denote the bandwidth, k the (true) number of expected clusters, ϵ and MinPts the neighborhood radius and the number of points defining a core point for DBSCAN, σ the standard deviation of the RBF kernel of spectral clustering.

Dataset		MS	BMS	SMS	DSMS	DBSCAN	k-means	Spectral
Aggregation	N	20	10	14.5	9.5	**5**	7	7
	K	0.89	0.93	0.91	0.98	0.94	0.88	**0.99**
	ACP	0.99	**0.99**	0.98	0.98	0.88	0.88	0.98
	ALP	0.79	0.88	0.83	0.98	**0.99**	0.88	0.99
Compound	N	41	24	29.4	18.9	**5**	6	6
	K	0.73	0.85	0.79	0.84	**0.92**	0.73	0.91
	ACP	**0.98**	0.97	0.97	0.95	0.93	0.76	0.92
	ALP	0.54	0.73	0.65	0.74	0.90	0.71	**0.91**
Spiral	N	73	36	34.1	30.8	**3**	3	3
	K	0.47	0.32	0.33	0.37	0.99	0.33	**1**
	ACP	**1**	**1**	**1**	**1**	**1**	0.33	**1**
	ALP	0.22	0.10	0.11	0.14	0.99	0.33	**1**
Jain	N	14	7	5.92	4.75	**3**	2	2
	K	0.45	0.55	0.70	0.72	0.88	0.76	**1**
	ACP	0.99	1	1	0.99	1	0.75	**1**
	ALP	0.21	0.30	0.49	0.53	0.77	0.78	**1**
A3[1]	N	29	19	24.4	21.2	**21**	21	21
	K	0.94	**0.98**	0.97	0.97	0.86	0.94	0.96
	ACP	**0.99**	0.97	0.96	0.96	0.87	0.94	0.96
	ALP	0.88	**0.99**	0.98	0.98	0.85	0.93	0.95
Iris	N	3	3	4.35	3.63	2	3	3
	K	0.85	0.88	0.88	**0.89**	0.73	0.82	0.88
	ACP	0.84	0.83	0.88	**0.89**	0.66	0.83	0.89
	ALP	0.85	0.92	0.88	**0.92**	0.81	0.83	0.87
Seeds	N	1.34	2	2.7	2.57	**3**	3	3
	K	0.63	0.80	0.82	0.82	0.79	0.82	**0.86**
	ACP	0.43	0.70	0.75	0.74	0.83	0.82	0.81
	ALP	0.97	0.91	0.90	0.91	0.75	0.81	**0.92**
Banknote	N	**2**	3	3.34	2.81	2	2	2
	K	0.78	0.76	0.70	0.68	**0.89**	0.54	0.86
	ACP	0.74	0.76	0.71	0.66	**0.84**	0.52	0.75
	ALP	0.82	0.75	0.68	0.70	0.95	0.56	**0.98**
Behavior	N	13	9	15.1	**7.35**	1	4	4
	K	0.84	0.82	0.82	**0.85**	0.64	0.47	0.73
	ACP	0.88	0.84	**0.89**	0.84	0.49	0.45	0.59
	ALP	0.80	0.81	0.75	0.86	0.84	0.48	**0.89**

Table 5: Comparison of all clustering methods across datasets, for the average number of clusters N and the metrics K, ACP, and ALP. For each criterion, the best result is displayed in bold, and the best results among all Mean-Shift based methods are underlined. For SMS and DSMS, the maximal number of iterations was set to 5.10^5. For k-means and Spectral Clustering, the exact number of clusters was provided as input in all cases.



(a) DBSCAN **(b)** DSMS (Proposed)

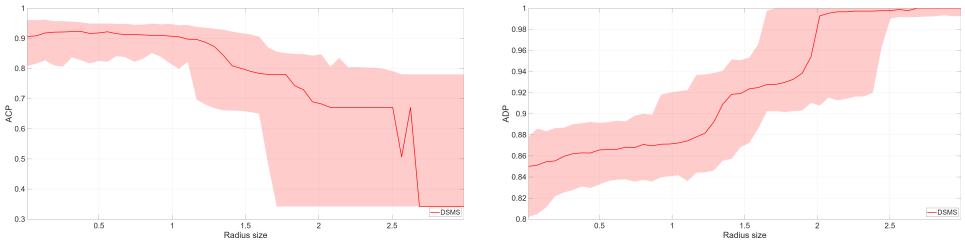


(c) K-means **(d)** Spectral Clustering

Figure 7: Clustering comparison on the Aggregation dataset (where clusters, though not multimodal, exhibit enough variability). DBSCAN fails on 'bridged' clusters, while K-means cannot model shapes. Spectral Clustering is highly accurate given $k = 7$. The proposed DSMS method correctly identifies all 7 ground-truth structures without requiring an a priori count.

Metric	**MS**	**BMS**	**SMS**	**DSMS**
N	112	55	278.066	61.1
K	0.9397	0.9660	0.9388	0.9674

Table 6: Comparison of all mean-shift based methods, for the average number of clusters N and the metrics K, on the Isolet dataset. For SMS and DSMS, the maximal number of iterations was set to 5.10^5.



(a) ACP **(b)** ALP

Figure 8: ACP and ALP as a function of $h_{max} - h_{min}$

Table 7: Average Elapsed Time of the MS-based approaches over 100 runs. Algorithms were stopped either when the shifts (for MS and BMS) and the average of the 1000 last shifts (for SMS and DSMS) are smaller than the user-defined tolerance $\varepsilon < 10^{-5}$, or when the maximal number of iterations (5.10^5) is reached.

Dataset Name	**Average Elapsed Name (s)**			
	MS	**BMS**	**SMS**	**DSMS**
Aggregation	0.1427	0.4156	22.7684	22.6349
A3	1.1919	4.2824	80.4458	90.5144
Compound	0.0606	0.0446	17.3128	18.1472
Jain	0.0625	0.0358	18.1175	16.9166
Spiral	0.0339	0.0733	16.8132	16.6884
Iris	0.0234	0.0114	14.9394	15.3166
Seeds	0.0402	0.0201	16.8300	17.1012
Banknote	1.6058	0.5312	30.0244	32.3458
Behavior	0.0004	0.0005	0.1720	0.1906

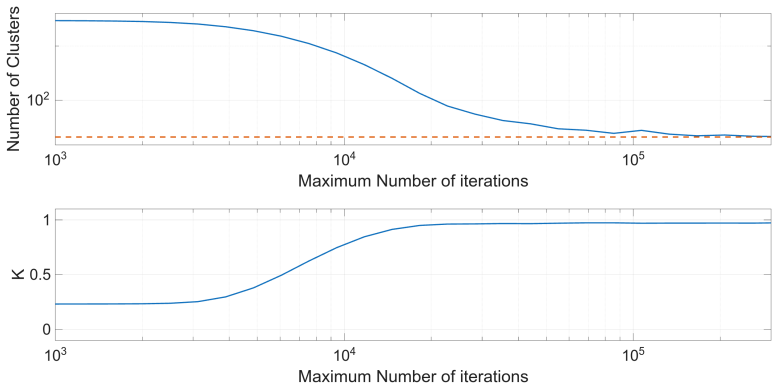


Figure 9: Evolution of the average number of clusters (upper figure) and average K (lower figure) over 100 runs of DSMS as a function of its maximal number of allowed iterations on the A3 dataset, limited to 21 clusters. The exact number of clusters is shown in dotted red in the upper figure.

Table 8: Average number of clusters N and average values of K for 50 DSMS executions on the Aggregation dataset, following different initializations of h_{min}, h_{max}, h_∞ and different rates of convergence.

Convergence Rate		Good setup ($h_{min}=1$, $h_{max}=4$, $h_\infty=2.5$)	h_∞ too large ($h_{min}=1$, $h_{max}=6$, $h_\infty=5$)	h_∞ too small ($h_{min}=0.5$, $h_{max}=4$, $h_\infty=1$)
$\nu_k \propto 1$	N	9.42	7.28	10.64
	K	0.99	0.98	0.98
$\nu_k \propto 1/k^2$	N	14.10	7.00	95.96
	K	0.91	0.95	0.56
$\nu_k \propto 1/\sqrt{k}$	N	14.80	7.02	89.78
	K	0.91	0.95	0.57
$\nu_k \propto 1/\log\log k$	N	9.5	8.67	27.10
	K	0.98	0.97	0.76